

MANAN KAPOOR

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EDUCATION

Thapar Institute of Engineering & Technology, Punjab, India 2024 – 2028
Bachelor of Engineering in Computer Engineering | CGPA: 8.19 / 10.00

EXPERIENCE

AI Engineering Intern, Colab91 Jun 2026 – Jul 2026
Gurugram, India | LLM Evaluation, Red-Teaming, Agent Memory Design, Prompt Caching, Full-Stack QA

- Ran **adversarial red-team evaluation** of a production LLM analytics agent across **50+ probes**, surfacing systematic failure modes: unsupported negative-filter queries answered as if satisfied, mislabelled result-set totals, self-contradictory aggregate counts, and confidently stated figures conflicting with the source snapshot.
- Diagnosed transcript-replay memory as the driver of context drift and per-turn cost growth in long sessions; specified a compact **factual-state store** hydrated from tool-computed values rather than model output, verified against **prompt-cache read/creation** metrics.
- Owned full-stack **QA** across the analytics dashboard, covering frontend, backend, and business-logic flows; documented defects in a structured tracker with priority, reproduction steps, direct page links, and screenshots for developer handoff.

Research Intern, Thapar Institute of Engineering & Technology Jan 2025 – Present
Patiala, India | NLP, Instruction Tuning, QLoRA, Dataset Engineering, Unsloth, HuggingFace

- Designed and published **PRISM-Punjabi** ([HuggingFace](https://huggingface.com)), the first multi-task Punjabi instruction dataset for controlled text rewriting: **16.43M tokens** from **600+ curated web sources**.
- Built a **Gemini 2.0 Flash** teacher-model supervision pipeline generating **91,000+ instruction-conditioned samples** across **6 rewriting tasks**; task-specific structural validation removed **37% corrupt samples**, reaching **100% instruction compliance** at **BERTScore 0.929+**.
- Full fine-tuned **LLaMA 3.1 8B** on **A100 GPUs** via **Unsloth** (Alpaca format, English instruction to Punjabi output); evaluated against **4 instruct baselines** across **5 metrics** on **6 tasks**.

PROJECTS

RAG Support Triage Agent github.com/manankapoor23/rag-support-triage-agent
Python, Sentence-Transformers, BM25, OpenAI API, Guardrails AI

- Built a RAG triage agent for 3 support ecosystems using hybrid dense+BM25 retrieval over a **117-doc** corpus, routing and answering **29 tickets** with zero external lookups.
- Implemented hard-grounding enforcement, forbidden-pattern filtering, and post-generation safety guardrails across a **16-module pipeline** with deterministic escalation.

PRISM: Multi-task Punjabi Instruction Dataset & Model huggingface.co/datasets/manankap23/PRISM-Punjabi
Python, PyTorch, Unsloth, HuggingFace, LLM Fine-Tuning, Gemini API, A100 GPU

- Published the first open multi-task Punjabi instruction dataset (**91,000+ samples**, CC-BY 4.0), filling a documented resource gap for a low-resource language.
- Benchmarked the fine-tuned **LLaMA 3.1 8B** against **4 instruction-tuned baselines** (LLaMA-3.1-Instruct, Mistral-7B, Gemma-7B, Phi-3-Mini) on BLEU, chrF++, ROUGE-L, BERTScore-F1, and Word-F1.

Natural Language to SQL Query System github.com/manankapoor23/NL-SQL
Python, LangChain, Gemini 2.0 Flash, SQLite, Slack Bolt, Prompt Engineering

- Built a Slack-native analytics assistant (`/ask-data` slash command over socket mode) converting plain-English questions into SQLite **SELECT** queries via **Gemini 2.0 Flash** orchestrated with **LangChain**, returning formatted result blocks in-channel.
- Grounded generation on an in-prompt table schema with sample rows at **temperature 0** for determinism, and enforced a **SELECT-only guardrail** rejecting all DDL/DML alongside markdown-fence stripping and a row cap before execution.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, R

ML & NLP: LLM fine-tuning (full and QLoRA), instruction tuning, RAG, transformers, benchmarking, prompt engineering

Frameworks: PyTorch, HuggingFace Transformers, Unsloth, LangChain, OpenAI SDK, sentence-transformers, scikit-learn, Pandas, NumPy

Backend & Data: FastAPI, Docker, model serving, inference pipelines, PostgreSQL, FAISS, Chroma, Git/GitHub, Linux